Presentation Title: IMAPS(Internet Message Access Protocol Secure)
Course Title: Computer Networks
Course Code:ICT-3201

Presented by :
Choity Barai
ID: IT22041
3rd year 2nd semester
Department of Information and Communication Technology, MBSTU

Presented to :
Dr. Nazrul Islam
Associate Professor,
Department of Information and Communication Technology, MBSTU

- **Table of indexes :**

- 1.Introduction to IMAPS
- 2.Purpose of IMAPS
- 3.Technical overview
- 4.IMAP Handshake
- 5.Issues of IMAP or Plain Email Protocols
- 6.How IMAPS works
- 7.Difference from Old Email Protocols (POP3, IMAP)
- 8.Features and Advantages of IMAPS
- 9.IMAPS Applications
- 10.Limitaions and challenges

Introduction to IMAPS:

IMAPS (Internet Message Access Protocol Secure) is a secure version of the IMAP protocol used for retrieving emails from a mail server. It provides **encrypted communication** between email clients and servers using **SSL/TLS**, ensuring that email data remains private and protected during transmission.

IMAPS allows users to **access and synchronize their emails across multiple devices** without removing them from the server, making it a reliable and secure solution for modern email communication.

Purpose of IMAPS:

The main purpose of **IMAPS (Internet Message Access Protocol Secure)** is to provide **secure, reliable, and convenient email access.**

1. Ensures **confidentiality** by encrypting email communication with **SSL/TLS**.

2. Allows users to **access and synchronize emails** across multiple devices.

3. Provides **reliable email retrieval** without removing messages from the server.

4. Protects emails from **interception and unauthorized access** during transmission.

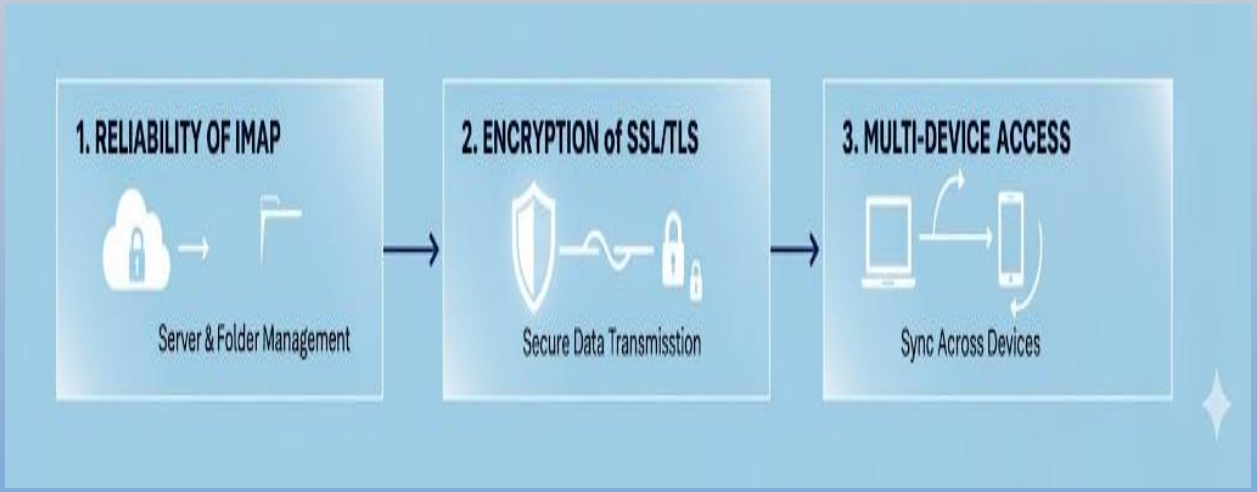
Technical Overview :

IMAPS is a **secure version of IMAP** that improves email retrieval by combining reliability, synchronization, and encryption:

1. Reliability of IMAP – keeps emails on the server and supports folder management.

2. Encryption of SSL/TLS – protects email data during transmission.

3. Multi-device access – syncs emails across devices without loss.



Handshake – Communication Flow

IMAP handshake is the initial communication between the **email client** and **IMAP server** to establish a **secure session**, perform **authentication**, and **sync mailbox data**.

Steps

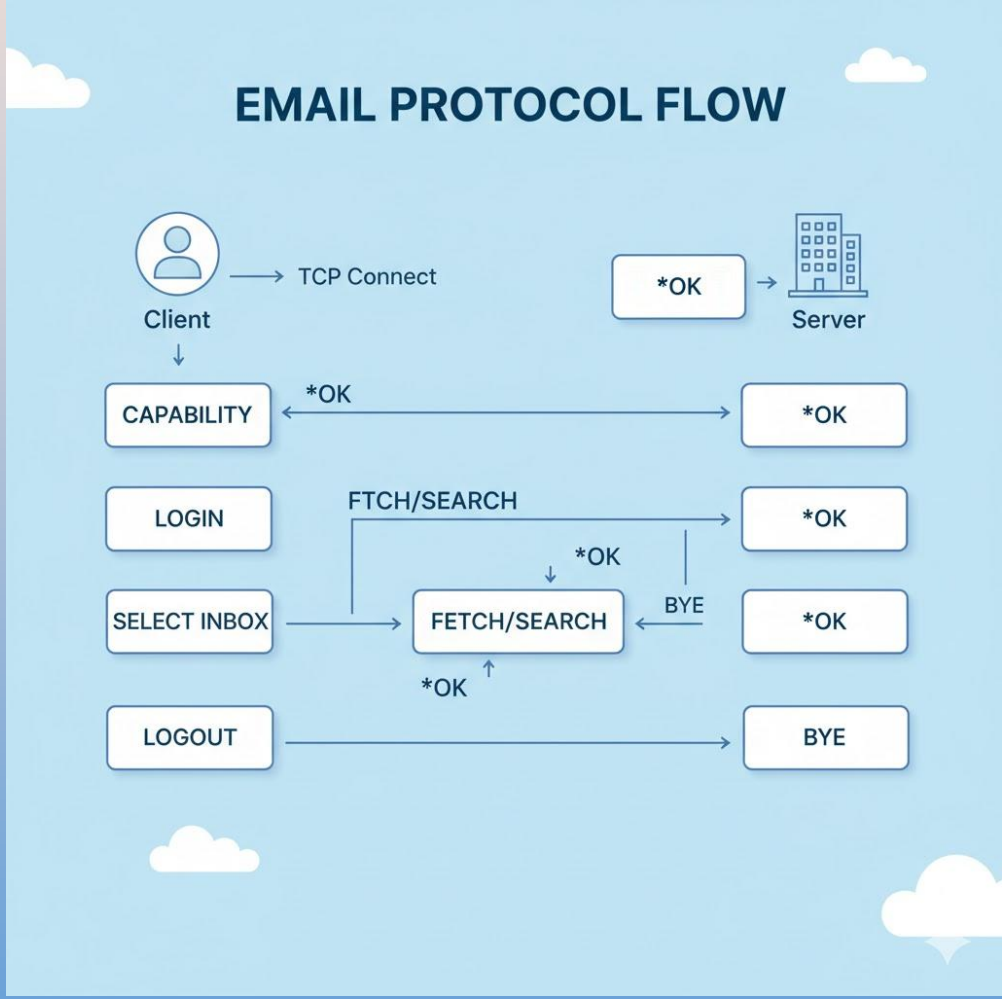
1. **TCP Connection** → Client connects to server (Port 143 = IMAP, 993 = IMAPS).
2. **Server Greeting** → * OK IMAP4rev1 Service Ready
3. **Client Capability Request** → A001 CAPABILITY
4. **Server Response** → Lists features (AUTH, IDLE, UIDPLUS).
5. **Authentication** → A002 LOGIN "user@example.com" "password"
6. **Mailbox Selection** → A003 SELECT INBOX
7. **Data Sync** → A004 FETCH / SEARCH
8. **Logout** → A005 LOGOUT

Secure Handshake (IMAPS)

- Uses **SSL/TLS** to encrypt all communication.
- Prevents eavesdropping & man-in-the-middle attacks.
- Ensures **data privacy** and **integrity**.

Flow Diagram

- Client → TCP Connect → Server
- Server → *OK
- Client → CAPABILITY → Server
- Client → LOGIN → OK
- Client → SELECT INBOX → OK
- Client → FETCH/SEARCH ↔ Server
- Client → LOGOUT → BYE



• Issues of IMAP or Plain Email Protocols

1. Security Issues

Plain IMAP sends data **without encryption (Port 143)**

- **No built-in data protection**

2. Synchronization Problems

- **Slow or inconsistent** updates across devices
- May cause **duplicate mails** or **delayed sync**

3. **Server Load & Storage**

- Keeps all emails on the **server** → increases load
- Large attachments reduce performance

4. Connectivity Dependence

- Needs **constant internet** connection
- Limited **offline access**

5.**Compatibility & Setup**

- Older clients lack **modern IMAP features**
- **Complex configuration** (ports, SSL, auth)

How IMAPS Works

IMAPS (Internet Message Access Protocol Secure) is the **encrypted version** of IMAP.
•It uses **SSL/TLS** to protect communication between the **email client** and the **mail server**.

Working Principle

1. Connection Setup

Client connects to the mail server on **Port 993 (IMAPS)**.
A **secure SSL/TLS handshake** is established before data transfer.

2.Authentication

User credentials are **encrypted** and verified by the server.
Prevents interception of usernames and passwords.

3.Mailbox Access

Client can **read, organize, or delete emails** directly on the server.
Messages stay on the server, so updates sync across all devices.

4.Data Transfer

Commands (like FETCH, STORE, SEARCH) are sent through the **secure channel**.
Both headers and message bodies are encrypted during transmission.

5.Synchronization

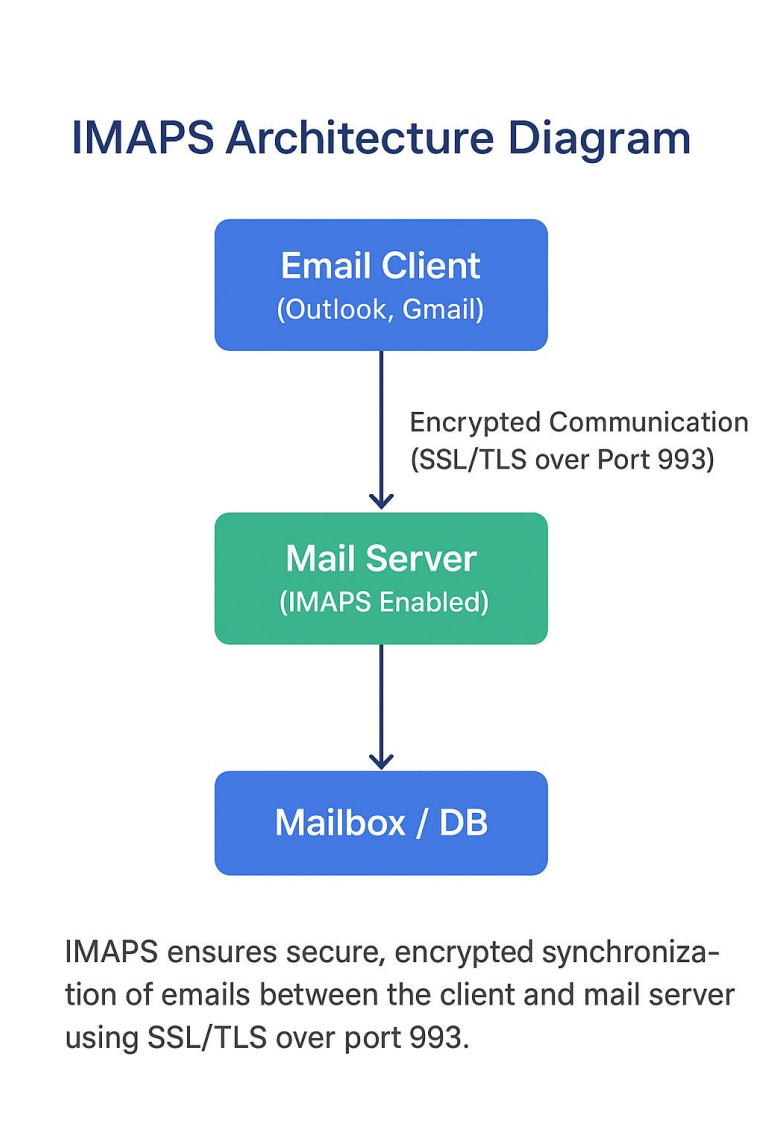
Any changes (read, delete, move) are **instantly reflected** on the server and all connected devices.

6.Session Termination

When finished, the client sends a **LOGOUT** command, closing the secure session.

IMAPS Architecture Diagram

- IMAPS client connects securely to the mail server using **SSL/TLS encryption**.
- Communication takes place over **port 993**.
- The **mail server stores all emails** centrally.
- The client **synchronizes** messages safely between mailbox and server.
- Ensures **confidentiality, integrity, and security** of data.



IMAPS vs SMTP

- **IMAPS** is used to **receive emails** securely from the mail server.
- **SMTP** is used to **send emails** from the client to the mail server or between servers.
- IMAPS works over **port 993**, while SMTP uses **port 465 or 587**.
- IMAPS is a **secure retrieval protocol**, and SMTP is a **secure transmission protocol**.
- Both use **SSL/TLS encryption** to ensure data privacy and protection during transfer.
- IMAPS handles **incoming mails**, whereas SMTP handles **outgoing mails**.

- **Difference from Old Email Protocols (POP3, IMAP)**

- POP3 downloads emails → removes from server.

- IMAP keeps emails on server → sync across devices.

- IMAPS = IMAP + SSL/TLS → **secure and encrypted**.

- Main advantage: Protects email data during transmission.

- **Features and Advantages of IMAPS**
- Secure email access via encryption (SSL/TLS)
- Syncs emails across multiple devices
- Download selected messages or headers only
- Supports folders, flags, and server-side search
- Protects emails from theft or tampering

IMAPS Applications

- **Corporate and Business Email**: Ensures secure communication within organizations.
- **Email Clients**: Used by desktop and mobile apps like Outlook, Thunderbird, and Gmail for syncing emails.
- **Webmail Services**: Enables secure access to emails through browsers.
- **Secure Alerts & Notifications**: Integrates with systems that require encrypted messaging.

Limitations and Challenges of IMAPS

Slower than POP3: Continuous server access can make it slower for offline use.

Requires Internet: Needs an active connection to sync emails across devices.

Server Storage Issues: Large volumes of emails can fill up server space quickly.

Configuration Complexity: May need advanced setup for firewalls, security, or corporate networks.

Future Scope / Improvements

- **Integration with modern OAuth2 authentication** for stronger, password-free login security.
- **Enhanced spam and phishing detection** to protect users from malicious emails.
- **Faster synchronization** for mobile and low-bandwidth networks.
- **Improved server-side filtering** for smarter inbox organization.
- **Better scalability and performance** to handle growing email traffic securely.

Thank you